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SS 17/7 PH® • UNS S17700

SS 17/7 PH® is a precipitation hardenable stainless steel that is formable and also capable of meeting high strength requirements. It is utilized in applications that require both high strength and good corrosion resistance. The alloy is classified as semi-austenitic to distinguish the soft phase (austenite) available for ease of forming from the high strength phase (martensite) usually achieved by heat treatment.

The alloy can be formed from the annealed temper but due to a rapid work hardening rate may not be deep drawn without intermediate annealing. It is joined by most arc and resistance methods used for stainless steel. It should not be fusion welded unless shielded by inert gas to prevent oxidation of aluminum.

Available Sizes:

SS 17/7 PH® is available from Hamilton Precision Metals as strip product in thicknesses from 0.001" to 0.040" (0.0254 mm to 1.016 mm) in widths to 12.0" (304.8 mm). The material conforms to ASTM A693, AMS 5528, AMS 5529, FED QQS766, and MIL S 25043.

Nominal Composition

SS 17/7 PH® NOMINAL COMPOSITION	
Chromium	16.8%
Nickel	7.3%
Alumium	1.2%
Carbon	0.08%
Iron	Balance

Mechanical Properties

SS 17/7 PH® MECHANICAL PROPERTIES				
	ANNEALED	ANNEALED HEAT TREATED	COLD ROLLED	COLD ROLLED HEAT TREATED
Ultimate Tensile Strength	130,000 PSI	235,000 PSI	220,000 PSI	265,000 PSI
Yield Strength (0.2% Offset)	40,000 PSI	220,000 PSI	200,000 PSI	260,000 PSI
Elongation in 2" *	35%	5%	2%	1%
Modulus of Elasticity (Tension)	29 x 10 ⁻⁶ PSI	-	-	-

*The measured elongation will be less as thickness decreases to 0.002" and less.
These values may be adjusted by control of process variables – consult HPM for desired values.

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Physical Properties

SS 17/7 PH® PHYSICAL PROPERTIES	
Density (Heat Treated)	0.276 lb/cu.in.
Electrical Resistivity @ R.T	83 Microhm·cm
Thermal Expansion Coefficient (Heat Treated) (21° to 93°C)	10.3 x 10 ⁻⁶ /°C
Thermal Conductivity @ 150°C	16.7 W/m·K
Magnetic Attraction	None

Typical values to guide alloy selection but are not a guarantee of minimum or maximum.

 Downloads

SS 17/7 PH® Strip and Foil Datasheet ([/-/media/ametekhpmetals/files/ss-17-7-ph.pdf?la=en&revision=ca377541-db50-44a1-855e-48f30784dda8&hash=CDA06CCB7191885A92F13D1909204](/media/ametekhpmetals/files/ss-17-7-ph.pdf?la=en&revision=ca377541-db50-44a1-855e-48f30784dda8&hash=CDA06CCB7191885A92F13D1909204))

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